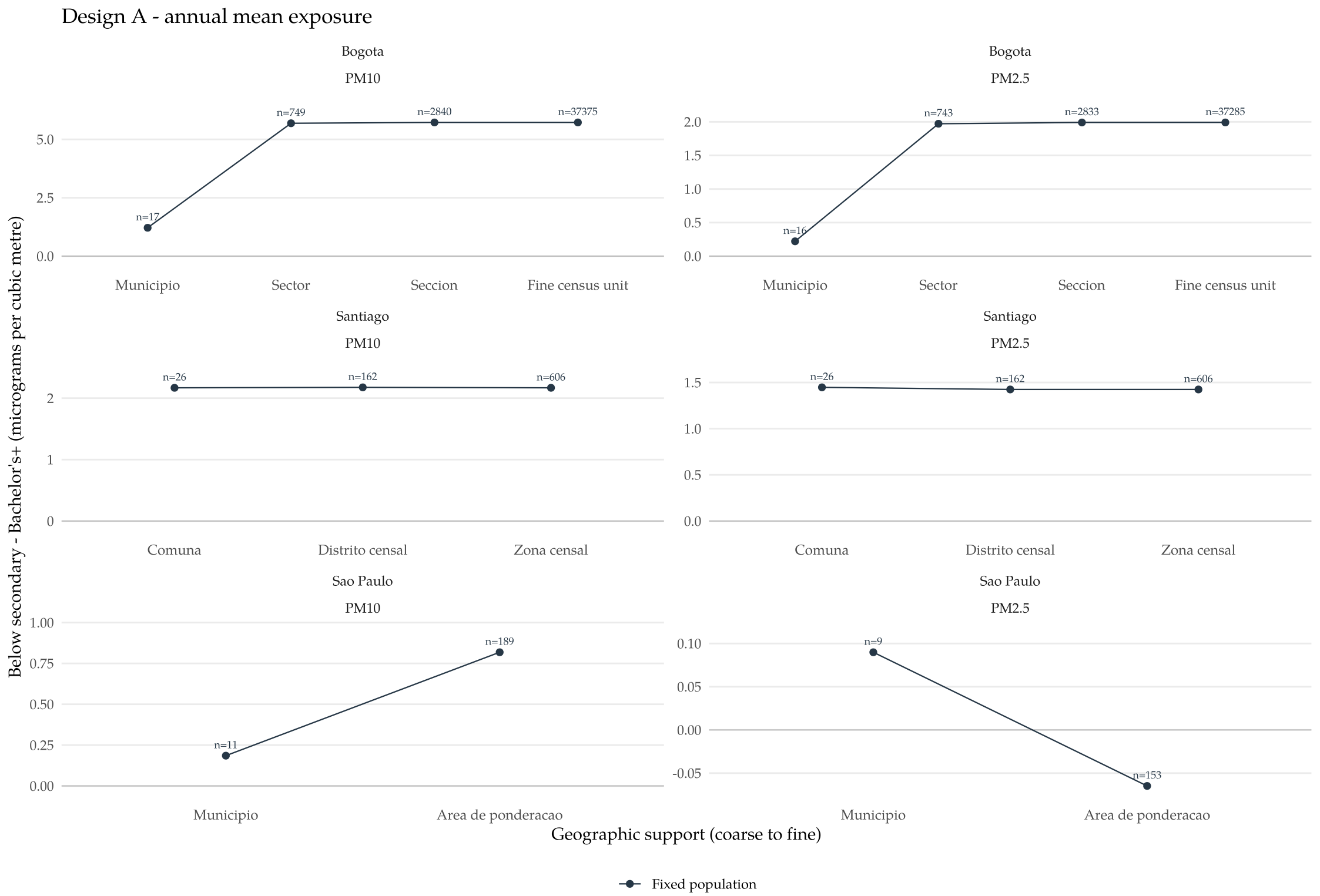
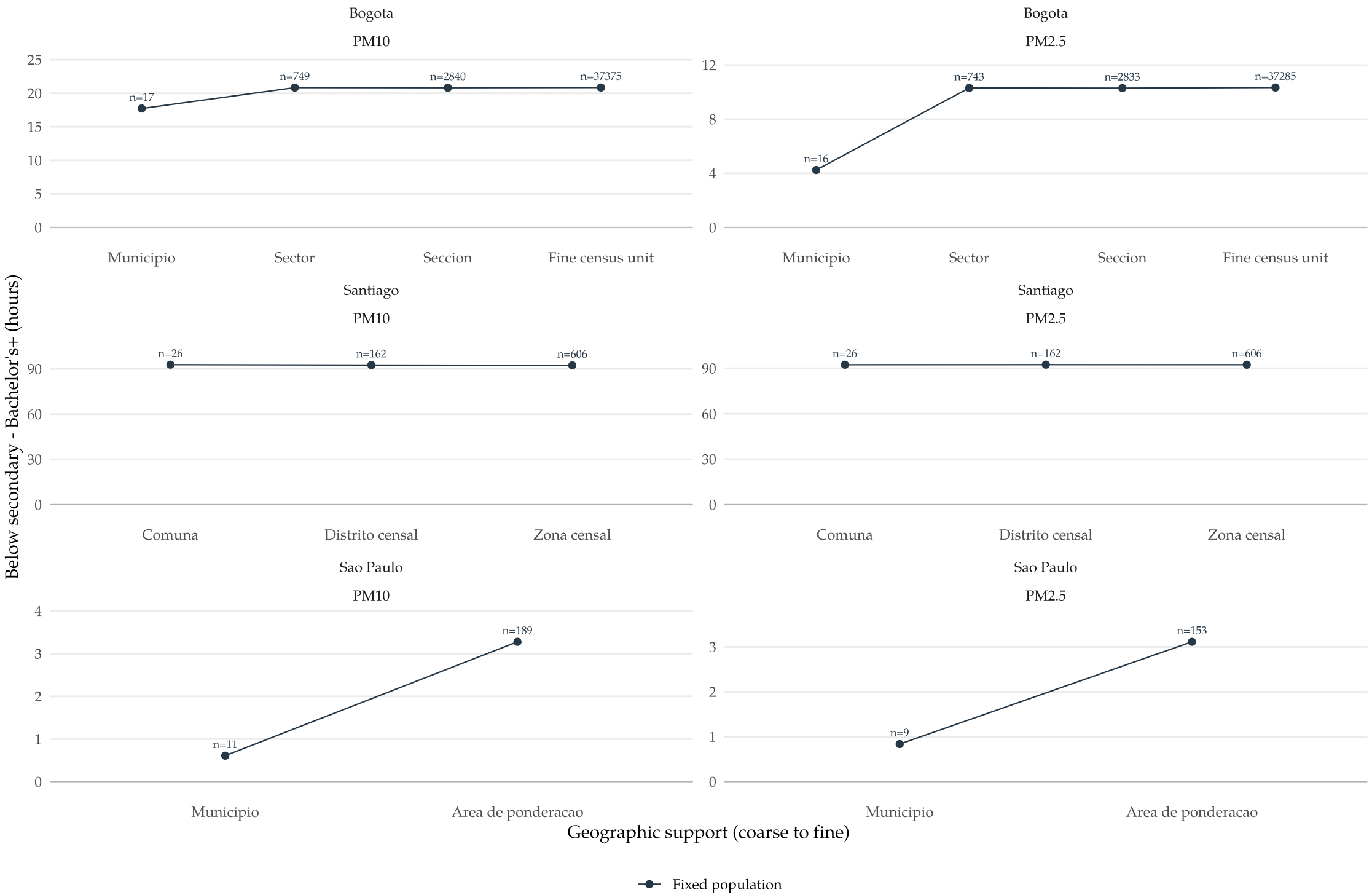


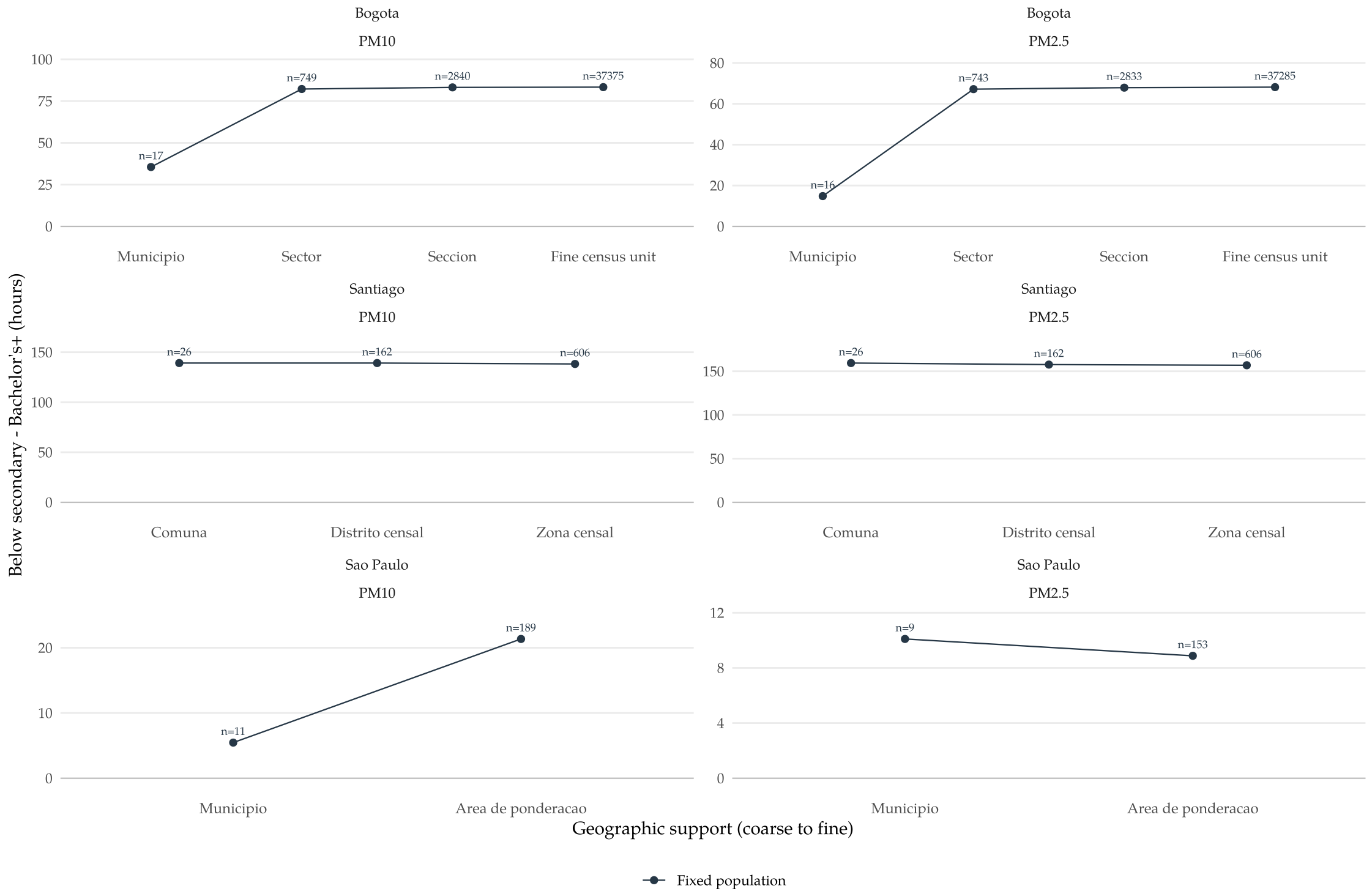
Fixed education-reporting adults and individual groups. Exposure aggregated with all-adult weights; locality branch shown separately.

Design A - annual hours at or above IT1



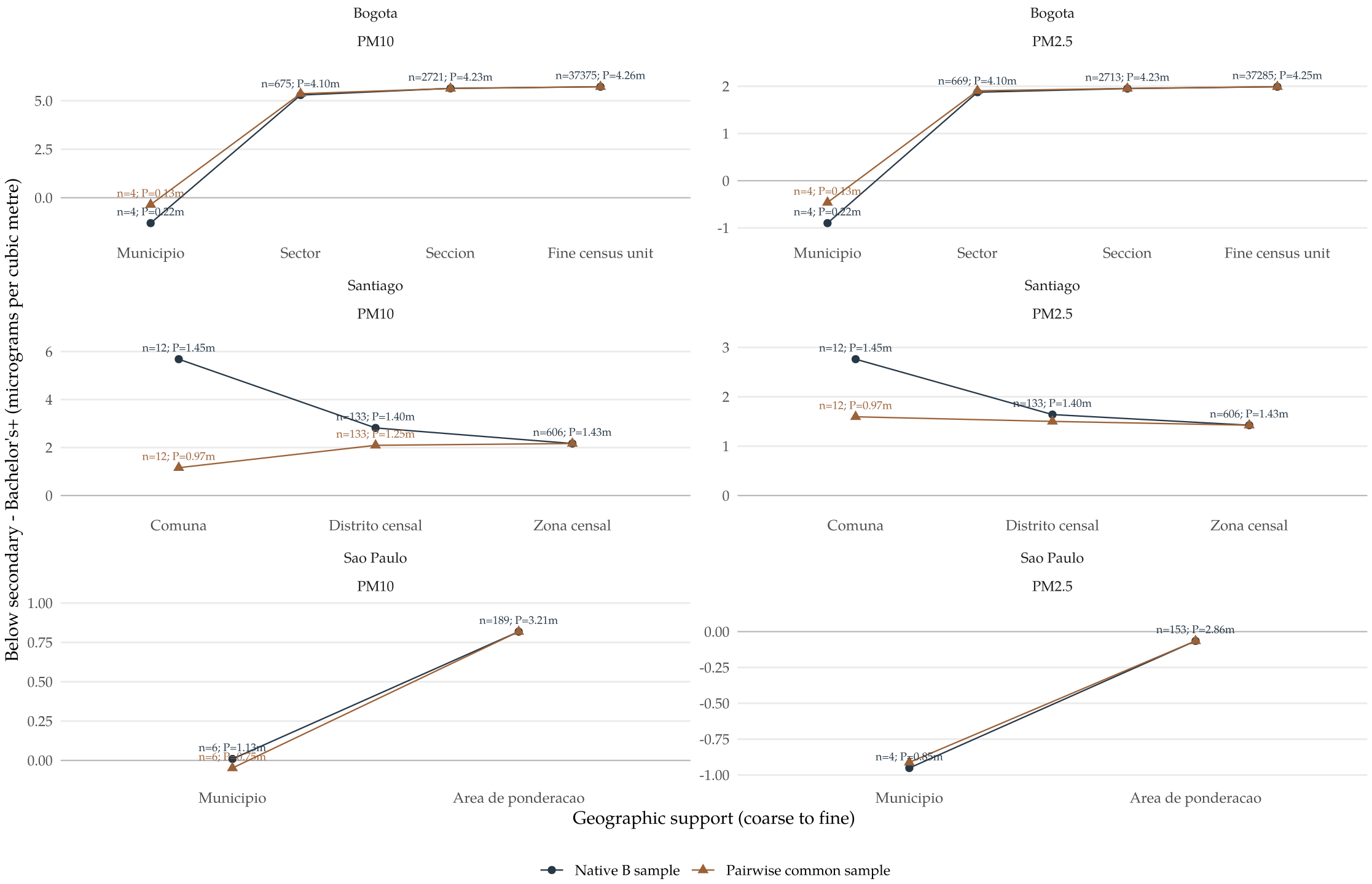
Fixed education-reporting adults and individual groups. Exposure aggregated with all-adult weights; locality branch shown separately.

Design A - annual hours at or above IT2



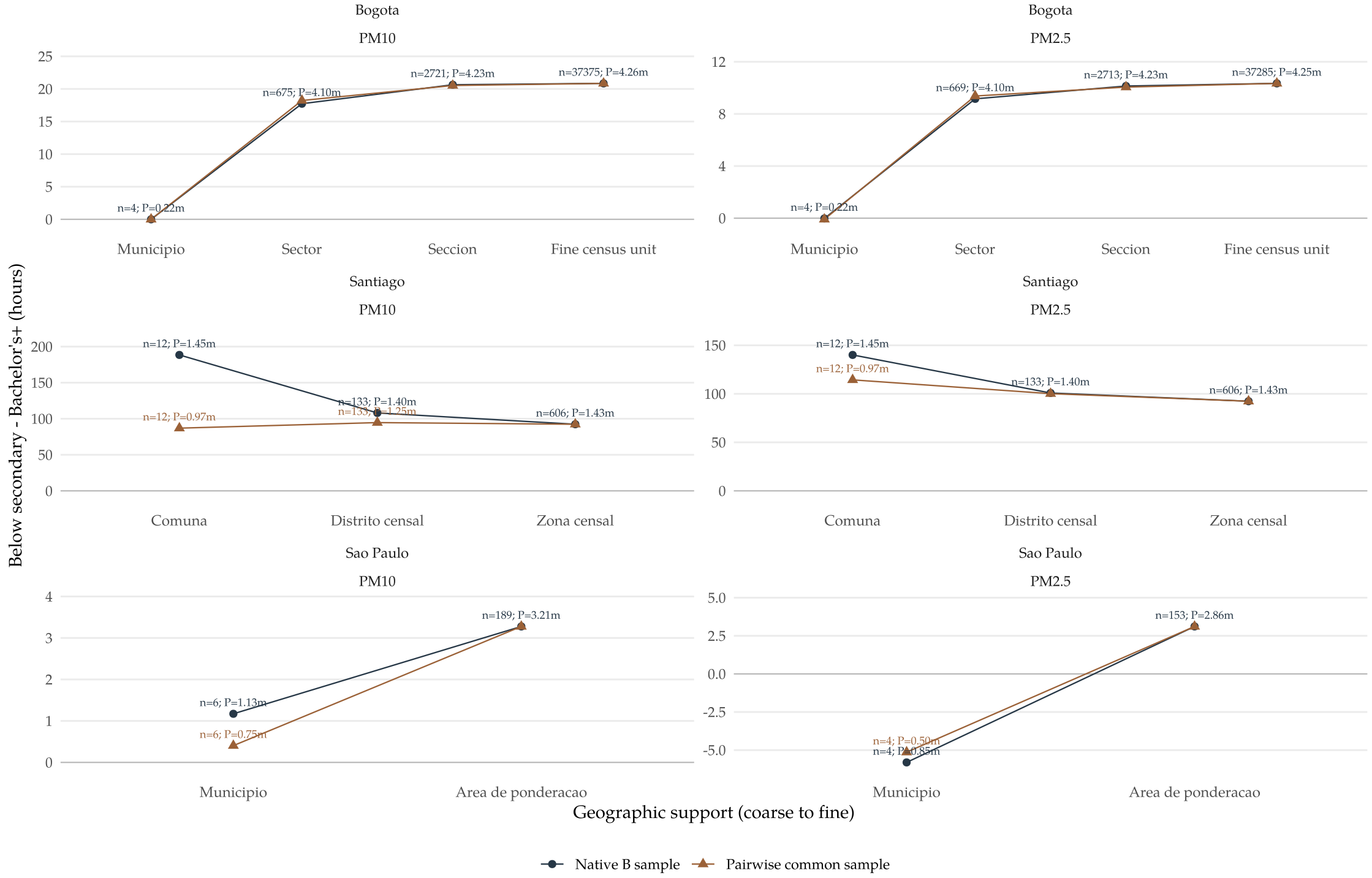
Fixed education-reporting adults and individual groups. Exposure aggregated with all-adult weights; locality branch shown separately.

Design B - annual mean exposure



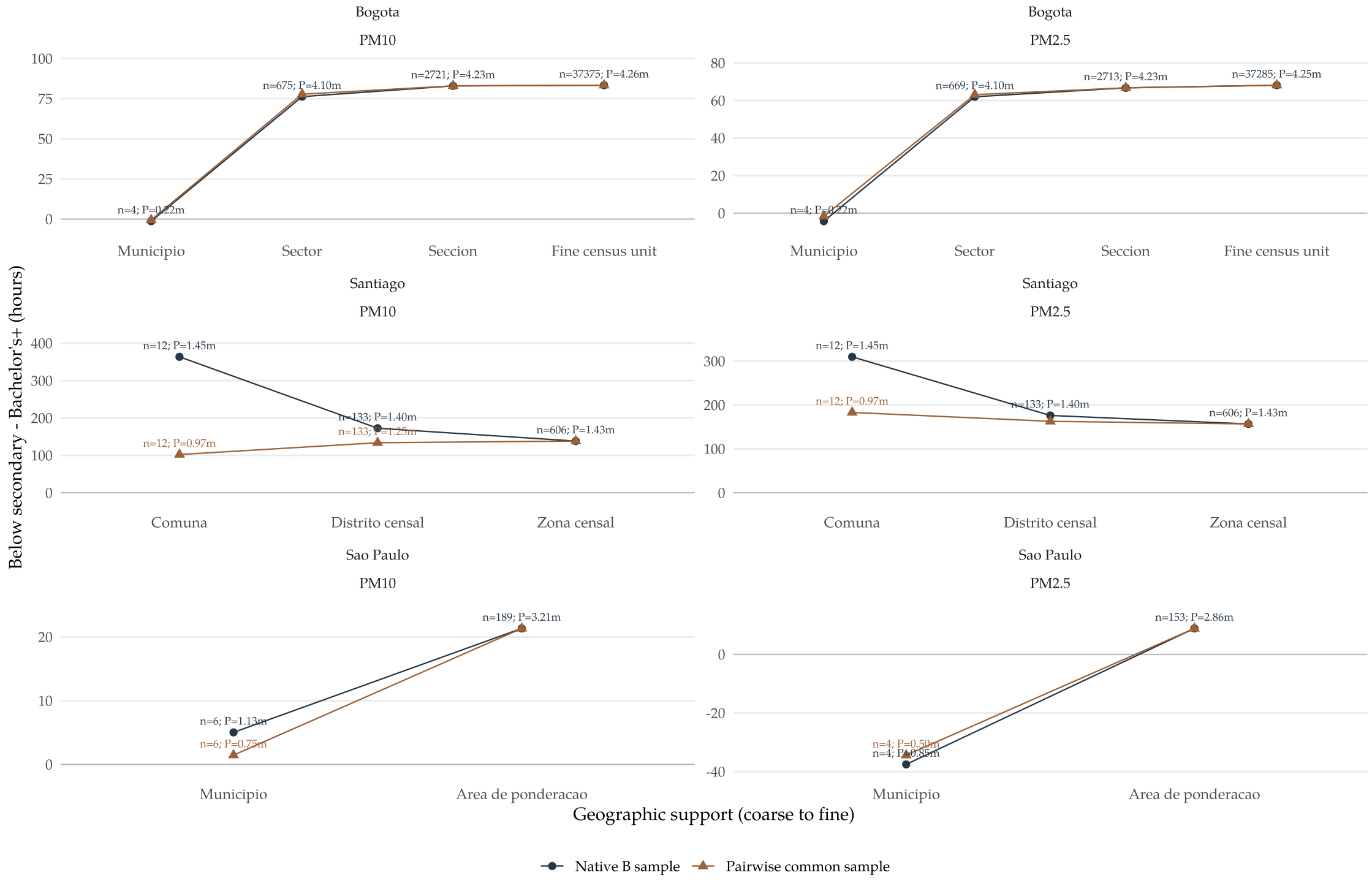
Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.

Design B - annual hours at or above IT1

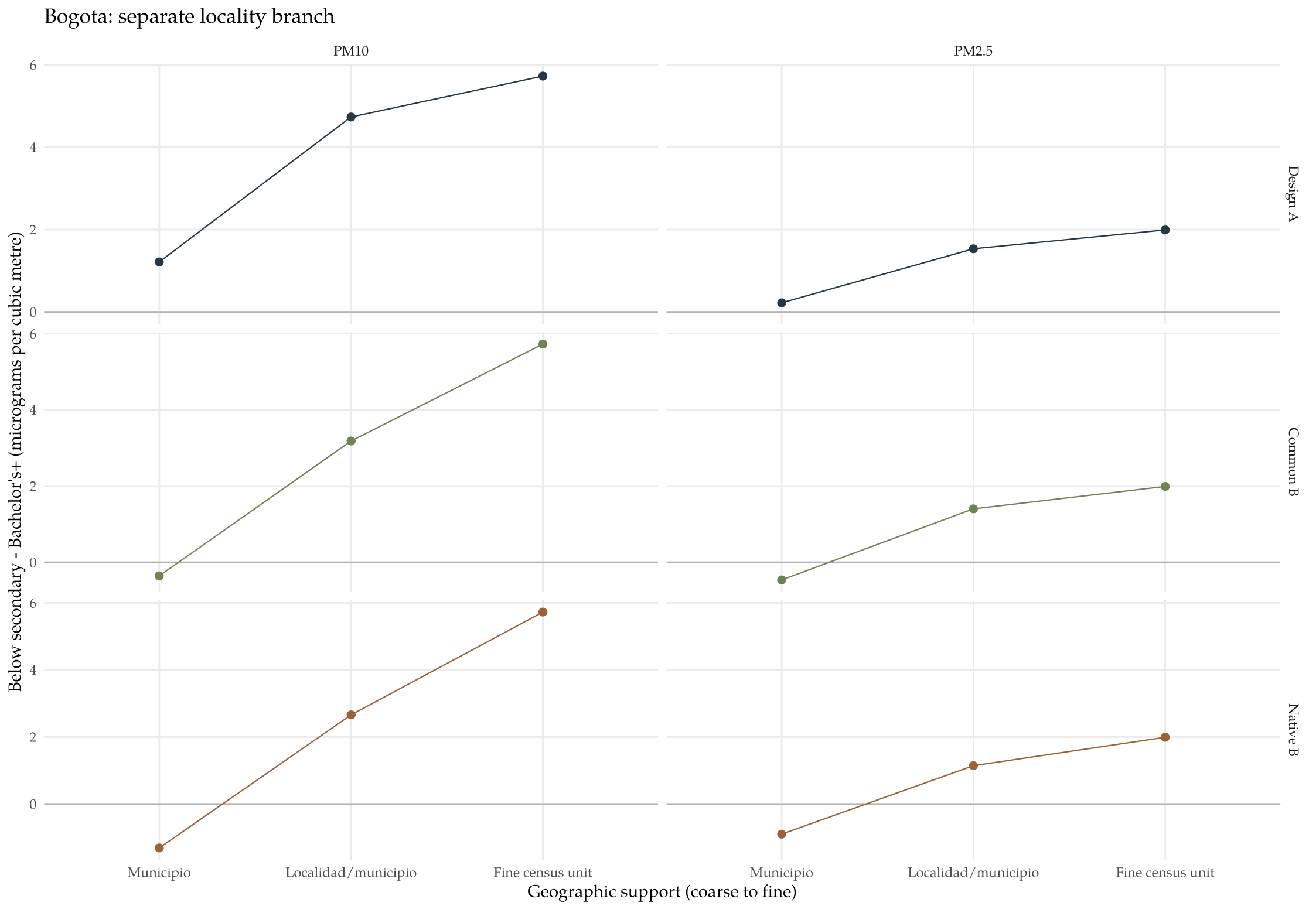


Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.

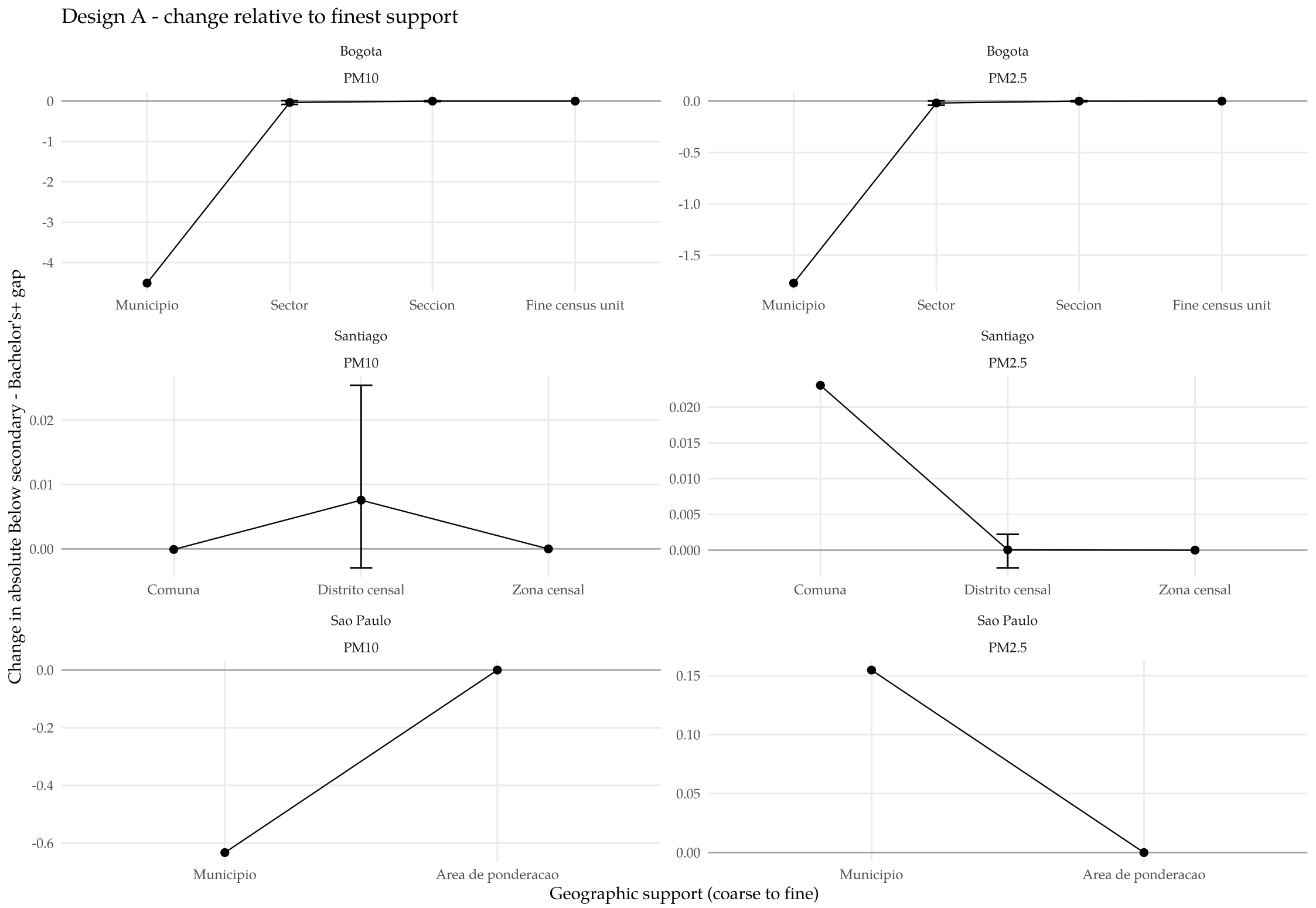
Design B - annual hours at or above IT2



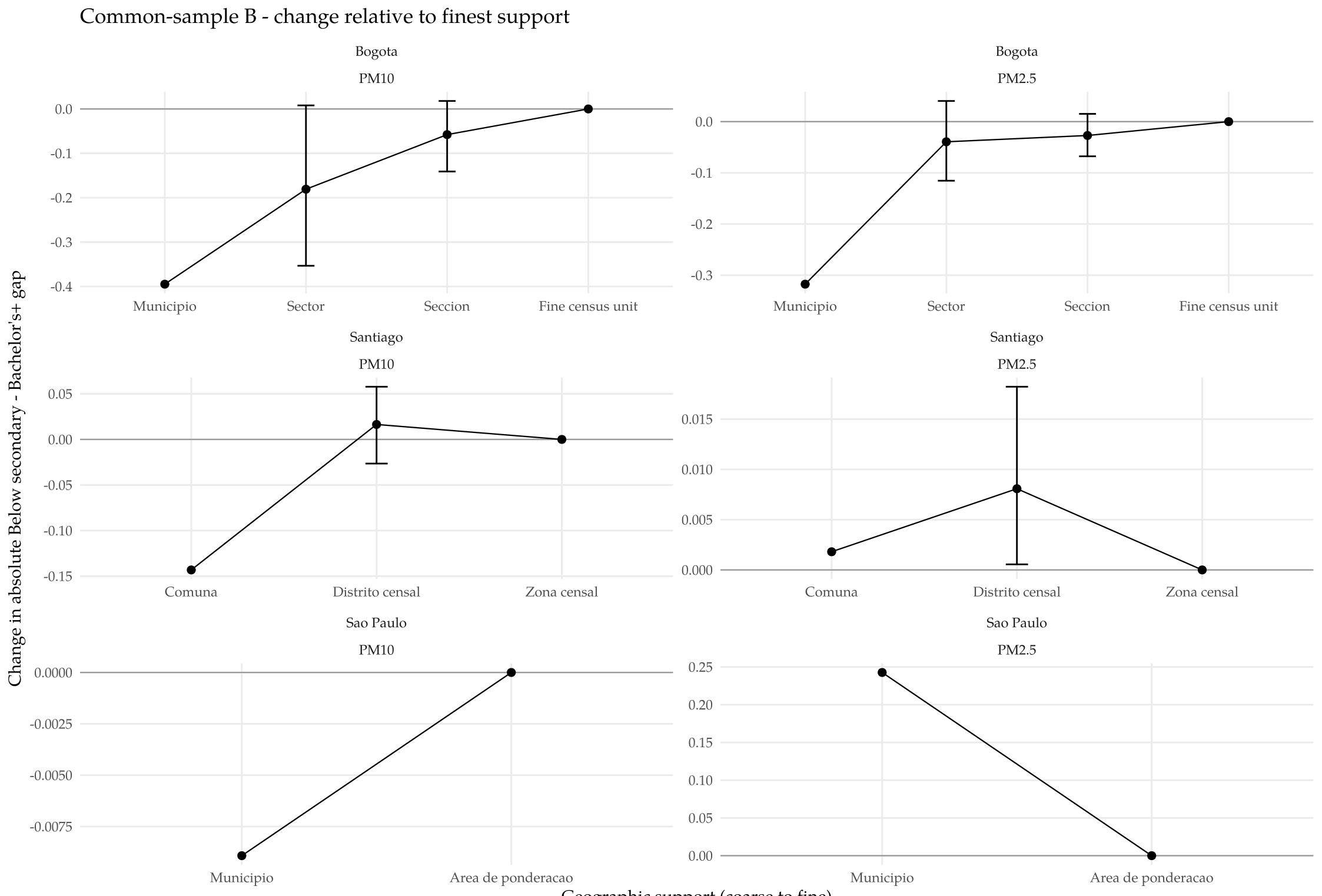
Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.



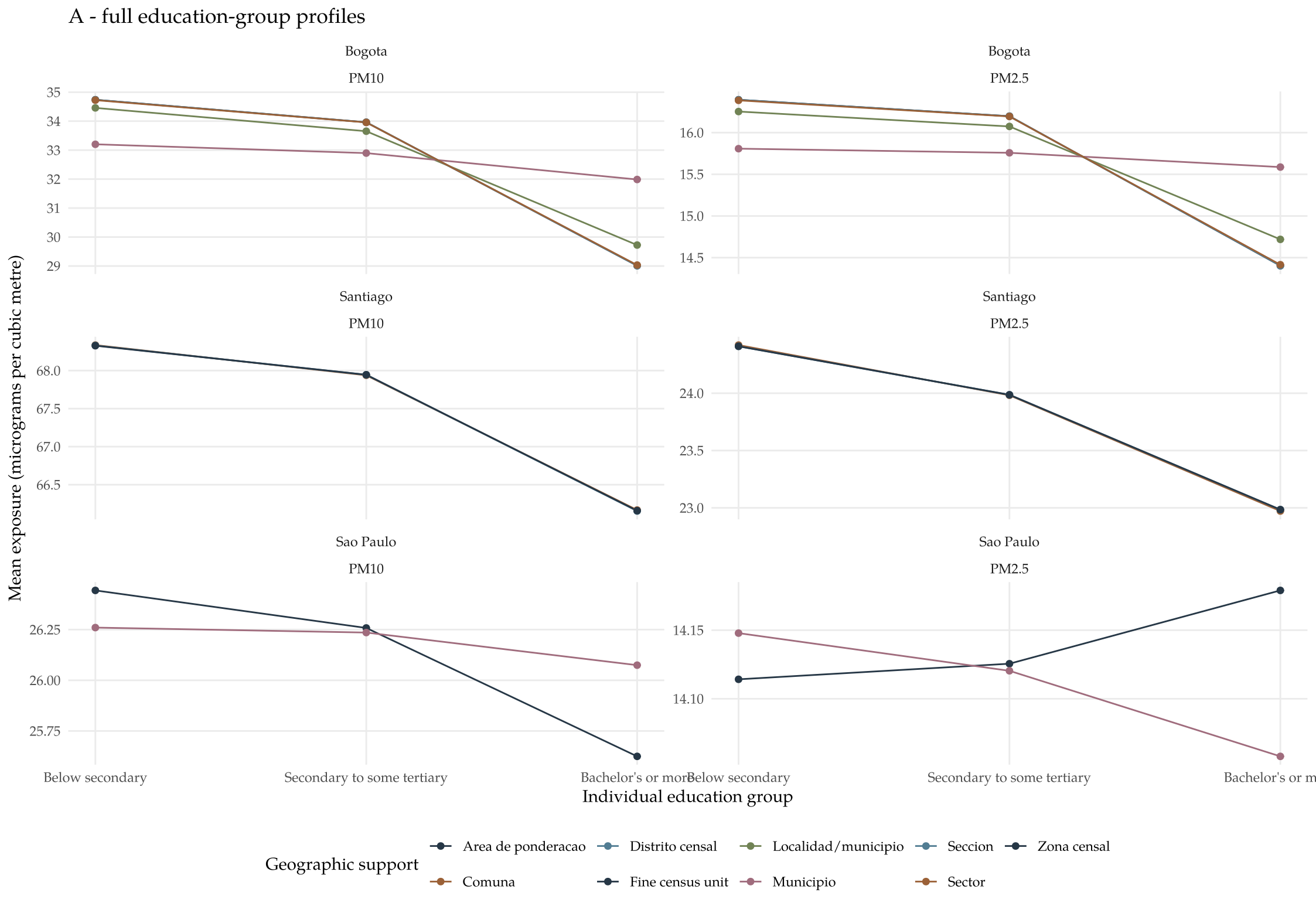
Descriptive comparisons.



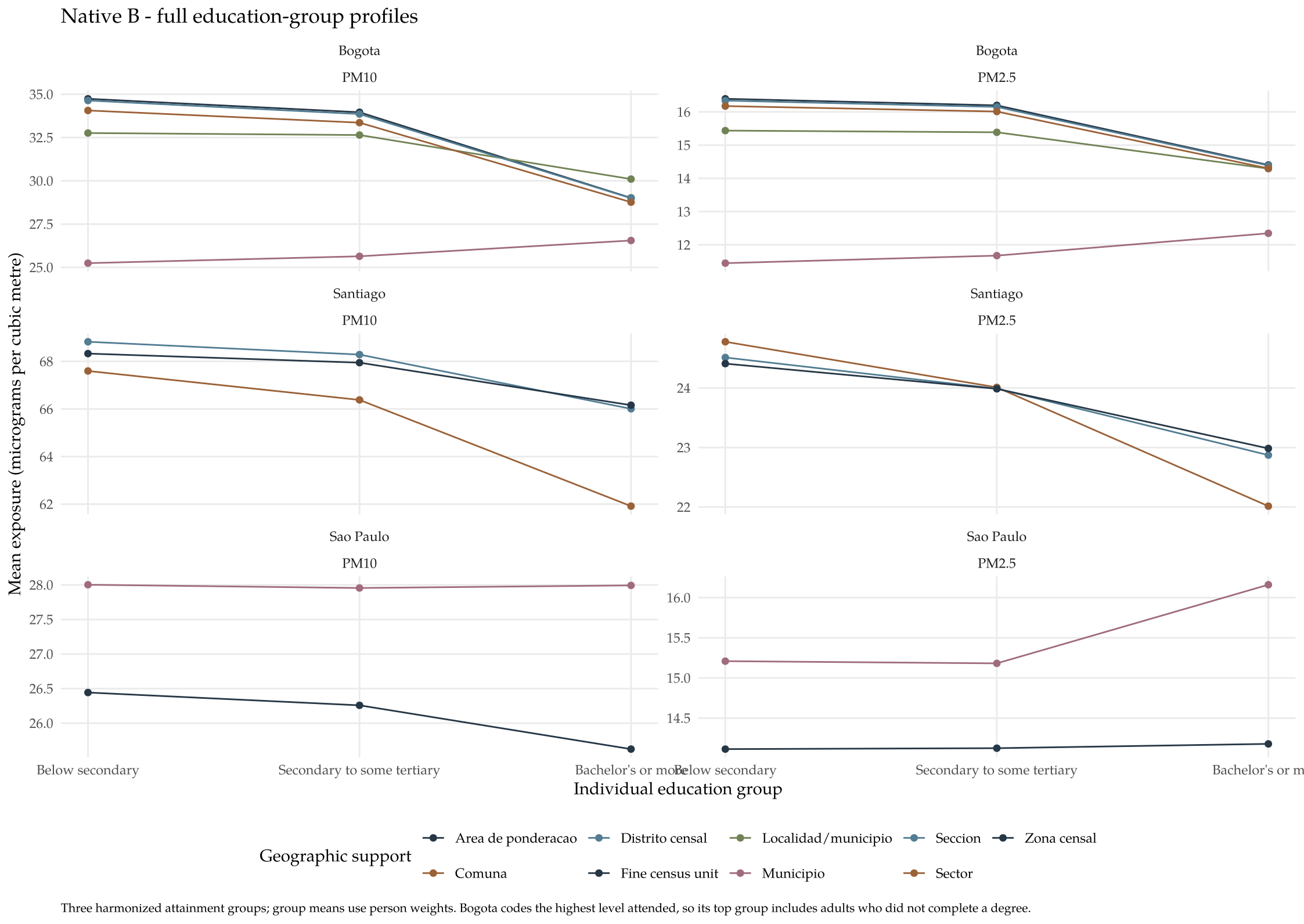
95% conditional geographic-cluster bootstrap intervals; 999 draws. Coarse comparisons are descriptive. B endpoints use the same pairwise population.



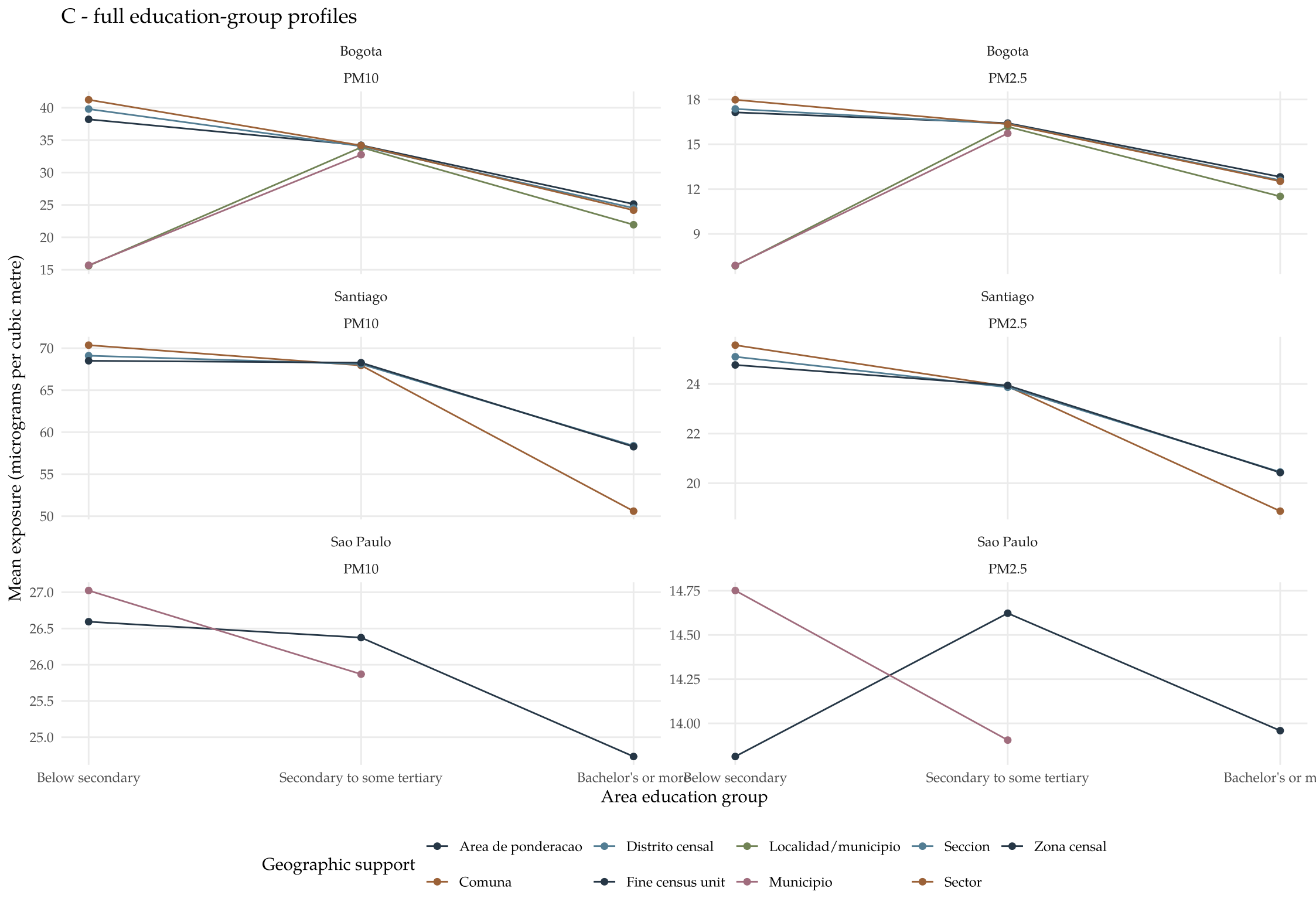
95% conditional geographic-cluster bootstrap intervals; 999 draws. Coarse comparisons are descriptive. B endpoints use the same pairwise population.



Three harmonized attainment groups; group means use person weights. Bogota codes the highest level attended, so its top group includes adults who did not complete a degree.

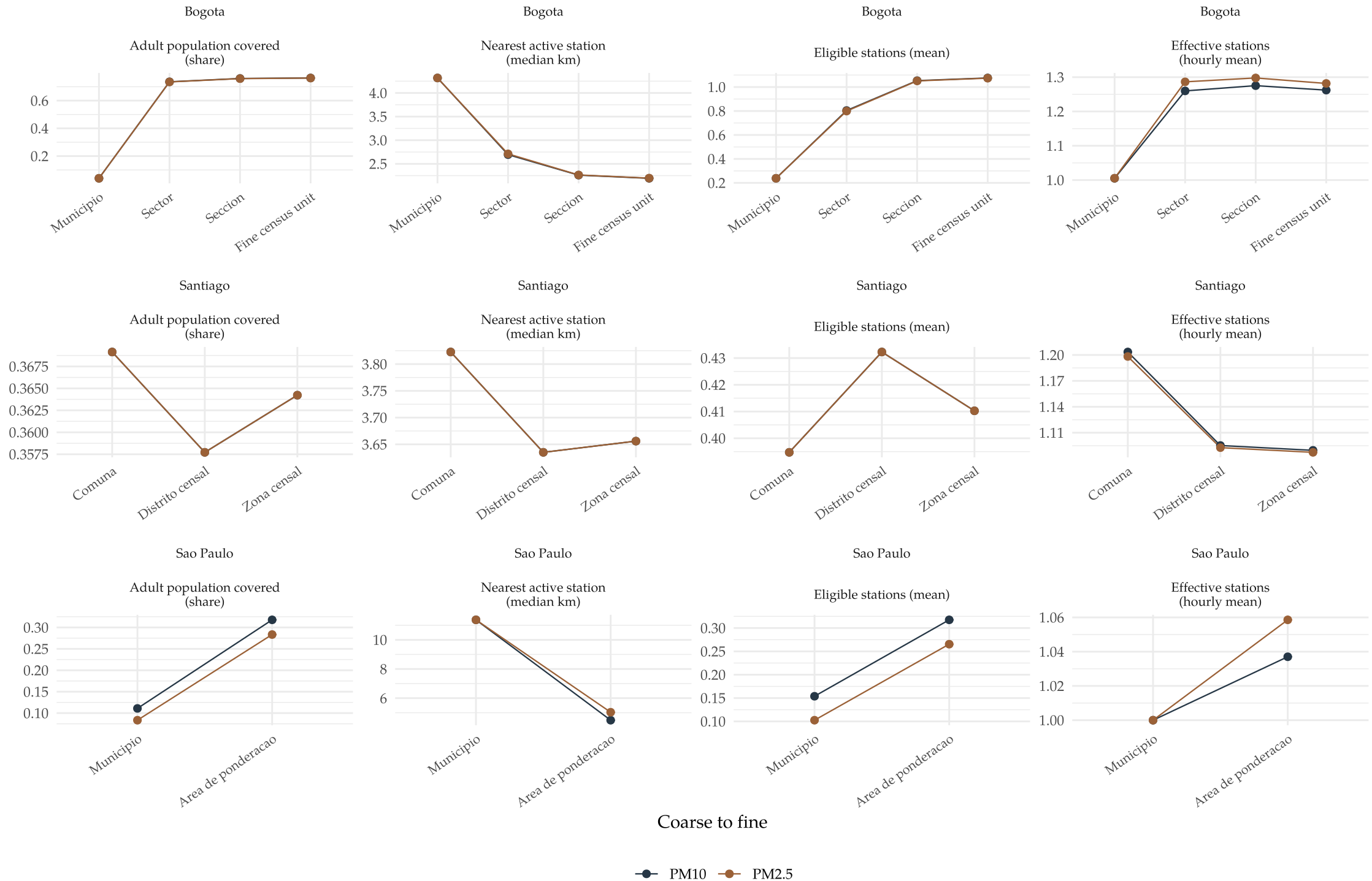


Three harmonized attainment groups; group means use person weights. Bogota codes the highest level attended, so its top group includes adults who did not complete a degree.



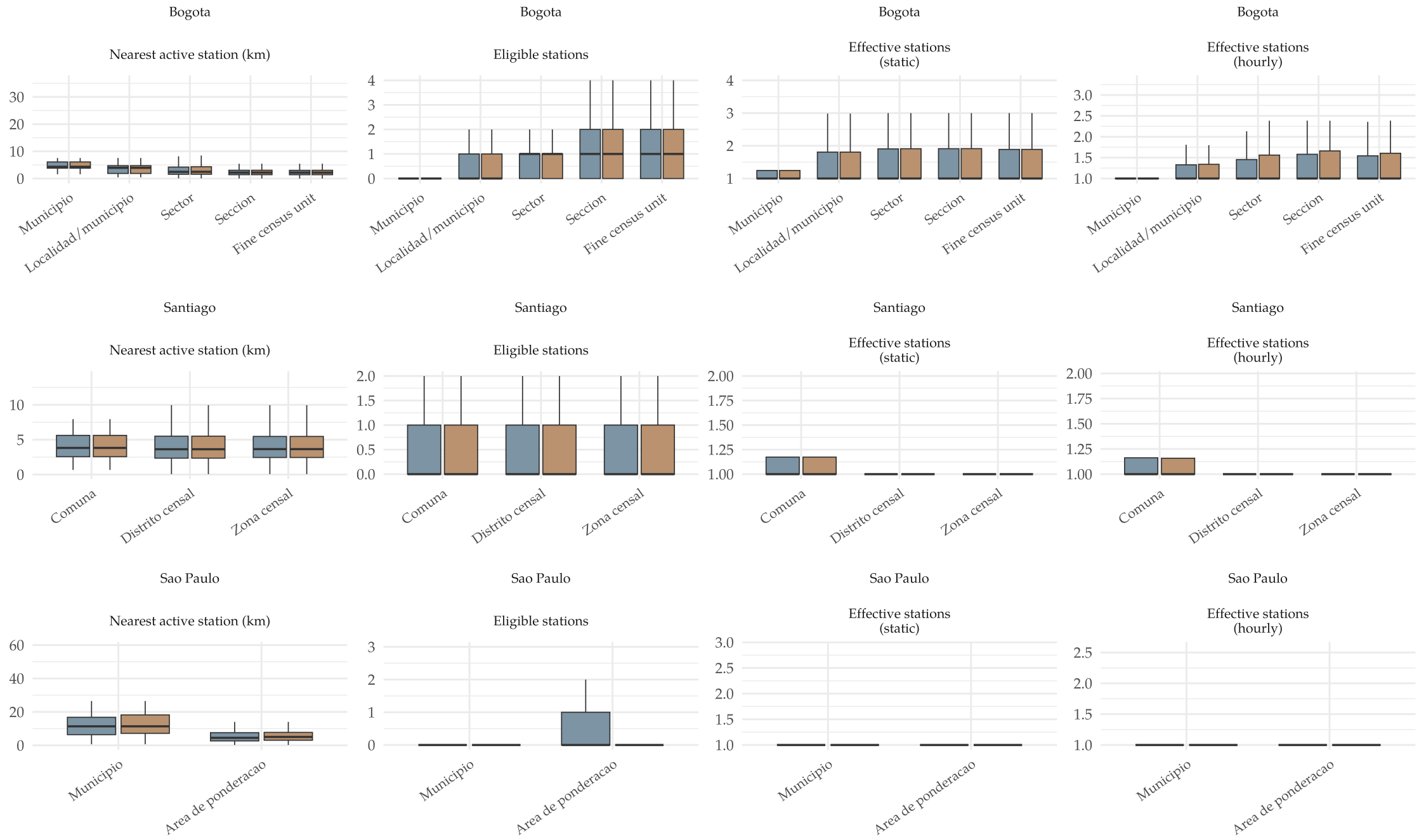
Classification only; finest exposure fixed. Empty area groups remain unavailable. These groups differ from A/B groups.

Monitoring-matrix sensitivity under Design B

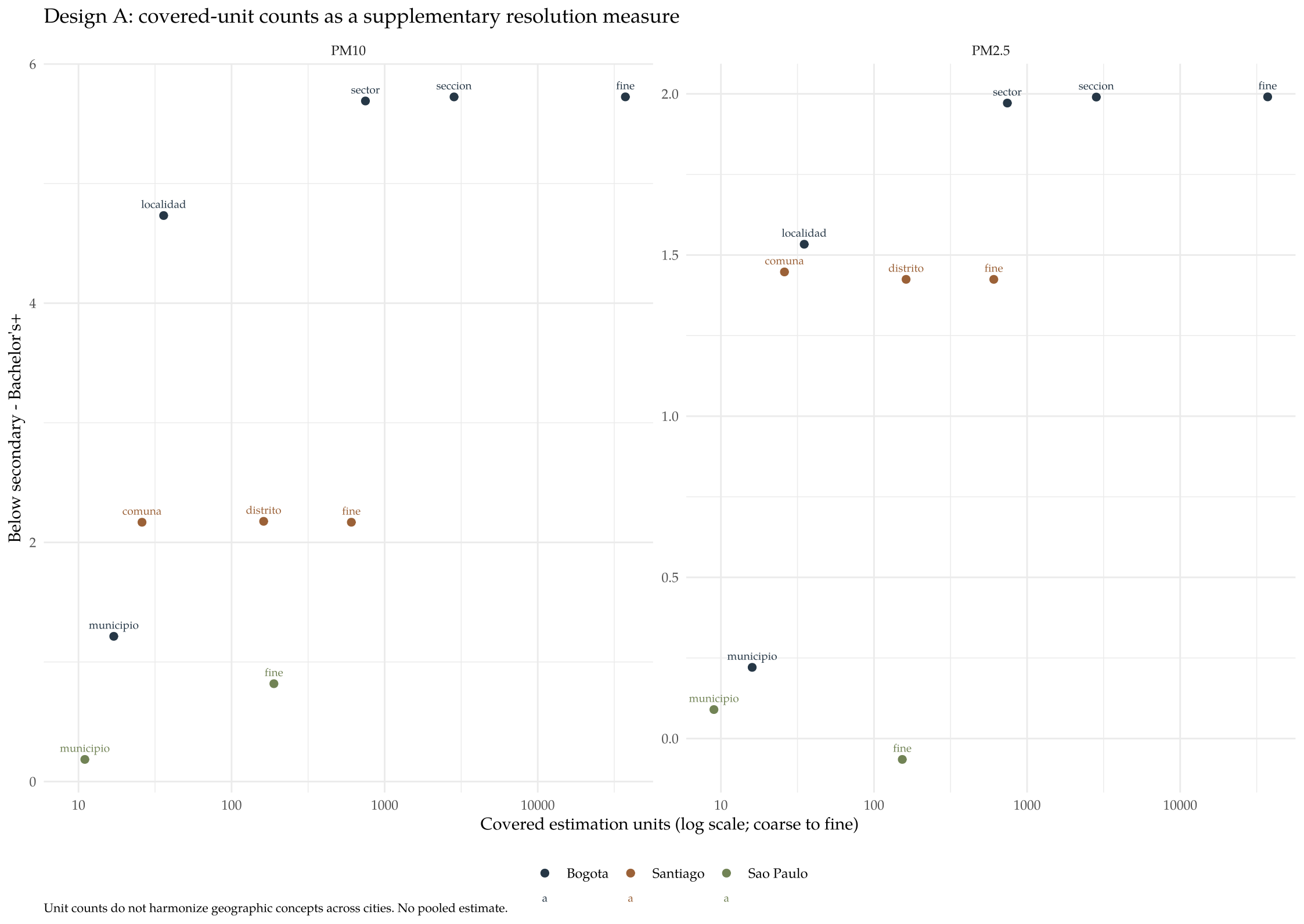


Catalog, active, eligible and contributing station counts are distinct in the tables.

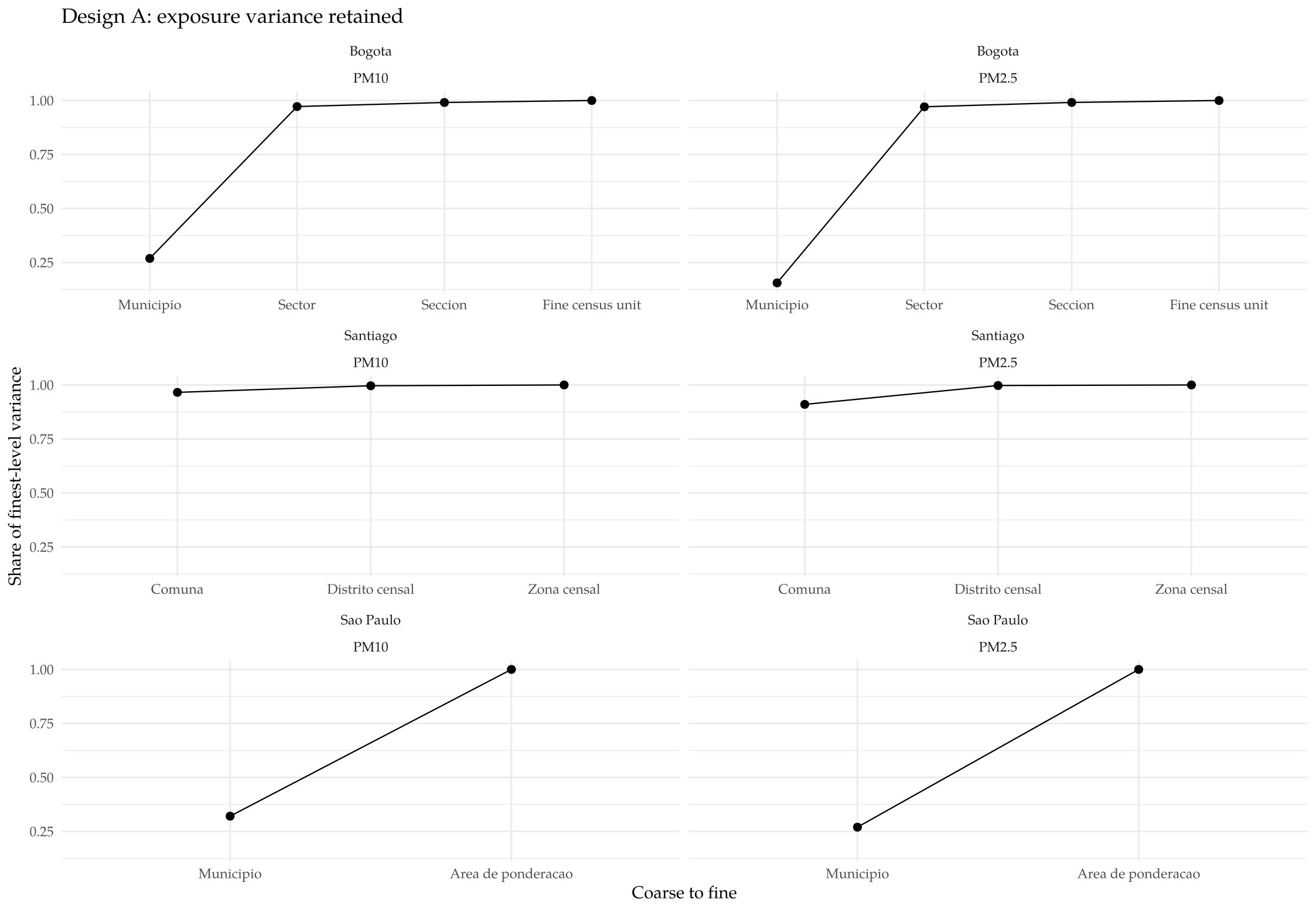
Monitoring-matrix distributions across populated units



Boxes: unweighted unit quartiles; whiskers: 1.5 IQR. Outliers omitted from view only.



Unit counts do not harmonize geographic concepts across cities. No pooled estimate.



All-adult aggregation weights and fixed covered support. Declining variance does not imply a monotone gap.